

# Huynh Le Dang Khoa

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## Research Profile

Honors Computer Science graduate from Ho Chi Minh City University of Technology (HCMUT – VNU-HCM), with research experience spanning **computer vision, foundation models, and embodied AI**. My current research direction is centered on **vision–language–action (VLA) models and robotic learning**, particularly long-horizon manipulation, memory-augmented policies, learning from human video, and scalable evaluation of embodied agents.

My previous work studies reliable visual reasoning under sparse, corrupted, and distribution-shifted observations, including monocular metric depth, anomaly detection, and egocentric hand–object perception. I am interested in connecting these perceptual capabilities with robot policies that must **perceive reliably, retain information over time, and act under changing real-world conditions**.

## Education

**Ho Chi Minh City University of Technology (HCMUT – VNU-HCM)** **Ho Chi Minh City, Vietnam**  
*B.Eng. Computer Science & Engineering — Honors Program* 2022 – 2026

- **Academic Standing:** GPA: **3.6/4.0**; Top **1%** of Faculty; degree classification: Excellent. Program completed; official diploma expected November 2026.
- **Undergraduate Thesis:** Accuracy-aware and failure-resilient scheduling for concurrent vision inference; grade: **9.84/10**.
- **Research Interests:** Vision–language–action models, robotic learning, embodied perception, multimodal foundation models, long-horizon memory, and robot evaluation.

## Research Experience

**Eye Code Tech / ARENA Lab, Singapore Management University** **Singapore / Remote**  
*Research Collaborator — Computer Vision & Robot Learning* *Feb 2026 – Present*

- **Vision Foundation Models:** Research on reliable visual reasoning with frozen or pretrained foundation models, including monocular depth grounding, acquisition-robust anomaly detection, and egocentric manipulation perception.
- **Embodied AI:** Developing research directions that connect visual perception to robot learning, with emphasis on memory-augmented VLA policies, video-derived demonstrations, and long-horizon manipulation.
- **Research Collaboration:** Collaborative research involving ARENA Lab at Singapore Management University and Do Van Minh's group at SMART (Singapore–MIT Alliance for Research and Technology).

**HPCC Lab, HCMUT** **Ho Chi Minh City, Vietnam**  
*Research Assistant — GPU Inference, Scheduling & Systems* *Jan 2025 – May 2026*

- **Efficient ML Inference:** Designed adaptive scheduling methods for concurrent heterogeneous vision workloads and evaluated accuracy–latency–throughput trade-offs under service-level constraints.
- **Thesis Results:** Achieved approximately **20% higher throughput** and **80% fewer SLA violations** than strong scheduling baselines on mixed ML inference workloads.
- **GPU Profiling:** Built reproducible inference benchmarks and profiled model latency, memory use, throughput, and scheduling behavior on modern NVIDIA GPUs.

**Dien Toan Group**

*AI Engineer Intern*

**Ho Chi Minh City, Vietnam**

*May 2025 – Aug 2025*

- **Retrieval-Augmented Generation:** Developed a document-understanding and RAG pipeline using multimodal language models, vector retrieval, PostgreSQL, MinIO, and Docker-based services.
- **Recognition:** Member of the team receiving the company’s **Most Outstanding Intern Team** award.

## Publications & Manuscripts

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*Reliable Few-Point Metric Grounding for Monocular Depth Foundation Models.* **ICLR 2027 submission, under review.** On NYUv2 with UniDepthV2, eight sparse metric observations reduce AbsRel from **0.0766** to **0.0540 (29.5% relative reduction)**; with 80% corrupted observations, AbsRel remains **0.0608**.

*Acquisition-Aware Anomaly Detection.* **Under review, Image and Computer Vision.** Introduces **ARAD**, a benchmark for anomaly detection under blur, motion, illumination, sensor noise, and mixed acquisition shifts, together with a training-free acquisition-aware detection framework.

*Mutual Hand–Object Reasoning for Egocentric Manipulation Perception.* **ICRA 2027 submission, under review.** Improves over CaRe-Ego by **6.93 mIoU points** on the EgoHOS out-of-distribution split and by **3.82 mIoU points** on mini-HOI4D in cross-dataset evaluation without fine-tuning.

*An Intelligent Virtual Machine Consolidation Approach for Energy-Efficient Cloud Data Centers.* **First author, ACOMP 2025.** [DOI](#)

## Selected Research Projects

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**MemoryPi — Memory-Augmented Vision–Language–Action Model** · Investigating explicit long-term memory for VLA policies. The policy stores and retrieves task-relevant information across subtasks so that actions can depend on observations or outcomes no longer visible in the current frame. Motivated by long-horizon failure modes observed in RoboMME-style manipulation tasks.

**Video-to-Simulation for Robot Learning** · Building a pipeline that uses computer-vision models to recover object geometry, hand–object interaction, and demonstrations from monocular human video, then retargets the reconstructed interaction into a robot simulation environment for learning and evaluation.

**Generative Evaluation for VLA Policies** · Exploring action-conditioned robot video generation and learned world models as a complementary evaluation signal for scenes that are difficult to reproduce with manually authored 3D assets. Current focus is understanding when predicted visual futures are sufficiently reliable to evaluate robot behavior.

## Technical Skills

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**Programming:** Python, C/C++, Java, Go, SQL, Linux, Git.

**Machine Learning:** PyTorch, OpenCV, CUDA; vision and multimodal foundation models; object detection and segmentation; monocular depth; VLA models; reinforcement learning; RAG and agentic AI.

**Robotics & Simulation:** Isaac Lab, robot simulation, manipulation evaluation, video-to-simulation, vision-based robot learning.

**Compute:** Access to NVIDIA **H100**, RTX **A5000**, and experience profiling and optimizing GPU inference workloads.

## Awards & Additional Information

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**Awards:** Most Outstanding Intern Team — Dien Toan Group; 2<sup>nd</sup> Place — Innostart Entrepreneurship Competition; Top 4 — UTE Junior Startup.

**Languages:** Vietnamese (Native); English (Professional Working Proficiency).